Equipment Maintenance Manual

MM-100 | Revision 7 | Effective 2026-01-15

Field	Value
Facility	Northfield Components Inc. - Plant 2 (Riverside)
Document owner	Maintenance Manager
Approved by	Maintenance Manager; Plant Engineer
Applies to	Maintenance technicians, setup technicians and floor supervisors at Plant 2
Review cycle	Every 12 months, or when equipment is added or modified

Note: FICTIONAL SAMPLE DOCUMENT - for software development and testing only. Names, equipment, part numbers and values are invented.

1. Scope and How to Use This Manual

This manual covers preventive maintenance (PM), routine servicing and first-line troubleshooting for Plant 2 production equipment. Each equipment chapter lists specifications, PM tasks and step-by-step procedures.

WARNING: Before any task that requires reaching into a machine, removing a guard or working on electrical, hydraulic or pneumatic parts, apply lockout/tagout as described in **EHS-SOP-001 Section 5**. Wear the PPE required by EHS-SOP-001 Section 4.2. If this manual and EHS-SOP-001 conflict, EHS-SOP-001 takes precedence.

"Operating hours" are read from each machine's hour meter. Plant 2 runs 3 shifts, 5 days a week, so **about 120 operating hours equal one calendar week**.

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2. Equipment Register

Equipment ID	Description	Location	Manufacturer / model (fictional)	Criticality
CNC-201, CNC-202, CNC-203	2-axis CNC turning centers	Line 1	Hartwell TL-250	A (bottleneck)
HP-400	400-ton hydraulic press	Line 2	Brannock HPX-400	A (bottleneck)
RW-510	Robotic MIG welding cell	Line 2	Kestrel RW-6 robot with Arcon 350 power source	B
CV-3	Belt conveyor, 30 m with 8 m incline	Line 3	Midland Belt MB-30	A (bottleneck)
CP-3	Automatic case packer	Line 3	Packworth CP-12	B
AC-1	75 hp rotary screw air compressor with 500 gal receiver	Compressor room	Torrent RS-75	A (plant-wide)
PW-1	Solvent parts washer, 30 gal	Parts-washing bay	CleanBay SW-30	C

Criticality A equipment stops the line or the plant when it fails; breakdowns must be responded to within 15 minutes. B: within 1 hour. C: within the same shift.

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3. Preventive Maintenance Schedule

Equipment	Task	Frequency	Ref.
CV-3	Visual belt tracking check	Every shift (by operator)	4.2
CV-3	Grease head, tail and take-up pulley bearings	Every 250 operating hours (about every 2 weeks)	4.1
CV-3	Check belt tension	Monthly	4.3
CV-3	Inspect drive gearbox oil level	Monthly	4.5
HP-400	Check hydraulic oil level and temperature; inspect for leaks	Daily	5.1
HP-400	Replace hydraulic return filter	Every 1,000 operating hours, or when the filter indicator shows red	5.3
HP-400	Hydraulic oil sample for lab analysis	Quarterly	5.2
HP-400	Change hydraulic oil	Every 4,000 operating hours, or when lab cleanliness is worse than ISO 18/16/13	5.2
HP-400	Lubricate gibs and slide	Weekly	5.1
CNC-201/202/203	Check way-lube reservoir; check coolant concentration and pH	Daily	6.1, 6.2
CNC-201/202/203	Grease chuck jaws	Weekly	6.4
CNC-201/202/203	Clean coolant tank and replace coolant	Every 6 months	6.2
RW-510	Replace contact tip; clean nozzle	Every shift	9.1
RW-510	Replace wire liner	Every 6 months	9.1
RW-510	Robot axis grease	Every 5,000 operating hours	9.2
AC-1	Verify receiver auto-drain operates; check oil level	Daily	7.1
AC-1	Replace air intake filter	Every 2,000 operating hours	7.2
AC-1	Change compressor oil and oil filter	Every 4,000 operating hours	7.2
PW-1	Replace solvent filter	Monthly	8.1
PW-1	Change solvent	Every 3 months, or sooner if the solvent is dark or parts come out oily	8.1
PW-1	Test lid fusible-link closure	Quarterly	8.2

4. Conveyor CV-3 (Line 3)

Specifications: 30 m total length including an 8 m incline at 15°; belt width 600 mm; belt speed 0.5 m/s (adjustable 0.3 to 0.8 m/s by VFD); drive motor 7.5 kW (10 hp) with helical gearbox; 6 flanged pillow-block bearings (head pulley 2, tail pulley 2, take-up pulley 2).

4.1 Bearing Lubrication

Interval: every 250 operating hours. Lubricant: NLGI grade 2 lithium-complex EP grease (stock code LUB-G2).

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1. Lock out CV-3 at disconnect DS-3C and clamp the incline belt (EHS-SOP-001 Section 5.3).
2. Wipe each grease fitting clean before connecting the grease gun.
3. Apply **2 pumps (about 3 g) per bearing**. Do not over-grease: excess grease pushes out the seals and causes overheating.
4. Wipe off purged grease, remove locks, run the conveyor for 10 minutes and check that bearing temperature stays below 70 °C (158 °F) with the infrared thermometer.
5. Record the task and bearing temperatures on the PM work order.

4.2 Belt Tracking

Operators check tracking every shift: the belt edge must stay at least 25 mm from the pulley frame. If the belt is drifting, notify maintenance. To adjust, lock out the conveyor, then turn the take-up adjusting screw on the side toward which the belt is drifting by a quarter turn, run for 2 minutes and re-check. Repeat until centered. Never adjust tracking with the guard removed while the belt is moving.

4.3 Belt Tension

With the conveyor locked out, press the return belt midway between two return idlers. Correct tension gives **10 mm of deflection per 1 m of span**. Adjust both take-up screws equally to keep tracking unchanged.

4.4 Clearing Jams and Guards

Never reach into CV-3 to clear a jam while it is running, even with the pull-cord stopped. Lock out at DS-3C first. Guards on CV-3 (pinch-point guards at head and tail pulleys and the incline side guards) may only be removed under lockout and must be reinstalled and checked before restart, per EHS-SOP-001 Section 6.

4.5 Gearbox

Check the oil sight glass monthly; it must be at mid-glass. Oil: ISO VG 220 synthetic gear oil (LUB-G220). Change every 10,000 operating hours.

5. Hydraulic Press HP-400 (Line 2)

Specifications: 400 tons capacity; stroke 400 mm; normal operating pressure 2,800 psi (193 bar), relief valve set at 3,000 psi (207 bar); reservoir 150 gal (568 L); hydraulic oil ISO VG 46 anti-wear (LUB-H46).

> **WARNING:** HP-400 has no minor-servicing exception. Every entry into the die area requires full lockout, accumulator bleed to 0 psi at valve HV-2, and both ram safety blocks in place (EHS-SOP-001 Section 5).

5.1 Daily and Weekly Checks

- Oil level must be between the MIN and MAX marks on the reservoir sight glass with the ram fully up.
- Normal oil temperature is 40-55 °C (104-131 °F). Above 60 °C, check the cooler fan and water flow. **At 65 °C (149 °F) the press must be stopped** until the cause is found.
- Look for leaks at hoses, fittings and cylinder seals. Any spraying leak means immediate shutdown: high-pressure oil can penetrate skin.
- Weekly: lubricate gibs and slide with way oil ISO VG 68 (LUB-W68) at the 8 lube points.

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5.2 Oil Sampling and Oil Change

Take an oil sample quarterly from the sample port SP-1 with the press at operating temperature and send it to the lab. Change the oil every 4,000 operating hours, or sooner if the lab reports cleanliness worse than ISO 18/16/13 or water above 500 ppm.

1. Run the press to warm the oil to about 40 °C, then lock out and bleed the accumulator.
2. Drain the reservoir into the waste-oil cart; remove the clean-out cover and wipe out sludge with lint-free cloths.
3. Replace the return filter (Section 5.3) and the breather cap.
4. Refill with new ISO VG 46 oil through a 10-micron filter cart; never pour directly from a drum.
5. Remove locks, cycle the press 10 times empty in inch mode to purge air, and top up the oil to the MAX mark.

5.3 Return Filter

Replace the return-line filter element (part FE-400-10, 10 micron) every 1,000 operating hours or when the differential pressure indicator shows red, whichever comes first.

5.4 Die Change

1. Lower the ram onto the die, lock out, bleed the accumulator and insert both ram safety blocks.
2. Unclamp and remove the die with the die cart; never slide dies by hand.
3. Install the new die, center it with the locating pins, and torque the die clamp bolts to **250 N·m (184 ft·lb)** in a cross pattern.
4. Remove blocks and locks, set shut height for the die per the die setup sheet, and run a first-article part for QC (QC-STD-200 Section 4).

6. CNC Turning Centers CNC-201, CNC-202, CNC-203 (Line 1)

Specifications: max turning diameter 250 mm; spindle up to 4,500 rpm; hydraulic chuck 8 in with 90 psi air-actuated tailstock; 60 gal coolant tank.

6.1 Daily Checks (operator)

- Way-lube reservoir above the MIN line; top up with ISO VG 68 way oil (LUB-W68).
- Coolant level, concentration and pH (Section 6.2).
- Chip conveyor running and chip bin less than three-quarters full.
- Door interlock check (EHS-SOP-001 Section 6.1).

6.2 Coolant

Coolant: semi-synthetic CoolCut 500 (LUB-C500). Target concentration **6-8%** measured with the refractometer (refractometer factor 1.0, so a Brix reading of 7 equals 7%). Target pH 8.5-9.5. If concentration is low, add premix at 10%; if high, add water. If pH is below 8.5 or the coolant smells sour, notify maintenance for treatment or replacement. Wear nitrile gloves and splash goggles when mixing coolant.

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6.3 Spindle Warm-Up

If the machine has been idle more than 4 hours, run warm-up program O9000 (10 minutes, stepping 500 / 1,500 / 3,000 rpm) before cutting production parts. Skipping warm-up causes thermal growth that can push the A-102 journal diameter out of tolerance.

6.4 Chuck Maintenance

Weekly, with the machine locked out: remove the jaws, clean the serrations and grease each jaw with chuck grease (LUB-CK) using 1 pump per fitting. Check chuck clamping pressure is 2.0-2.4 MPa on the machine gauge.

7. Air Compressor AC-1

Specifications: 75 hp rotary screw; output 300 cfm; cut-in pressure 95 psi, cut-out 110 psi; 500 gal receiver with electronic auto-drain; plant distribution set to 90 psi.

7.1 Daily Checks

- Press the TEST button on the receiver auto-drain and confirm water discharges.
- Oil level visible in the sight glass while running.
- Discharge temperature below 105 °C (221 °F). Above this, check the cooler for dust blockage.

7.2 Scheduled Service

Replace the air intake filter (FE-AC1-A) every 2,000 operating hours. Change the oil (synthetic compressor fluid LUB-RS8) and oil filter (FE-AC1-O) every 4,000 operating hours. Replace the air-oil separator every 8,000 hours. Before service, lock out DS-AC1, close the outlet valve and drain the receiver to 0 psi.

8. Solvent Parts Washer PW-1

8.1 Solvent and Filter

Replace the solvent filter (FE-PW1) monthly. Change the SafeClean 150 solvent every 3 months, or sooner if it is dark or parts come out oily. Pump spent solvent into the red hazardous waste drum. Follow the PPE in EHS-SOP-001 Section 4.2 (respirator required for solvent change).

8.2 Fire Lid

Quarterly, check the fusible link and lid closure: release the lid prop and confirm the lid falls fully closed. Replace a fusible link that is painted, corroded or has been exposed to heat (part FL-165, 165 °F rating).

9. Robotic Welding Cell RW-510 (Line 2)

9.1 Torch Consumables

Each shift: replace the contact tip (0.045 in / 1.2 mm, part CT-045) and clean spatter from the nozzle with the reamer station. Every 6 months: replace the wire liner. Enter the cell only with the robot locked out or in teach mode (EHS-SOP-001 Section 4.2 and 5.3).

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9.2 Robot Mechanical

Grease all six axis gearboxes with robot grease LUB-RG every 5,000 operating hours, following the robot manufacturer's purge procedure. After any crash, run the TCP check program P900 before restarting production.

10. Troubleshooting Guide

Equipment	Symptom	Likely causes	First actions
CV-3	Belt drifting to one side	Take-up uneven; material buildup on pulley; worn idler	Lock out; clean pulleys; adjust tracking (4.2).
CV-3	Bearing hot (above 70 °C) or noisy	Over- or under-greasing; misalignment; worn bearing	Check grease history; check alignment; replace bearing if above 80 °C or rough.
CV-3	Belt slipping at drive	Low tension; worn lagging; overload	Check tension (4.3); inspect lagging.
HP-400	Oil temperature above 60 °C	Cooler fan failed; low water flow; low oil level; relief valve bypassing	Check fan and water; check level; stop press at 65 °C.
HP-400	Press slow or not reaching tonnage	Low oil; clogged filter; pump wear; relief valve set low	Check oil and filter indicator; call hydraulic specialist if not resolved.
HP-400	Light curtain faults repeatedly	Dirty lens; misalignment; damaged cable	Clean lenses; check alignment LEDs; do not run until resolved.
CNC lathes	A-102 journal diameter drifting large	Tool wear; thermal growth (no warm-up); offset error	Check insert; confirm warm-up ran; adjust wear offset and notify QC.
CNC lathes	Poor surface finish	Worn insert; low coolant concentration; chatter	Replace insert; check coolant (6.2); check chuck pressure.
AC-1	Plant air pressure below 85 psi	Air leak; intake filter clogged; auto-drain stuck open	Check filter indicator; check drain; walk the system for leaks.
RW-510	Porosity in welds	Low shielding gas flow; dirty nozzle; drafts	Check gas flow 35-40 CFH; clean nozzle; check for open doors or fans.

11. Torque Specifications

Application	Fastener	Torque
HP-400 die clamp bolts	M20 class 10.9	250 N·m (184 ft·lb)
CV-3 pillow-block bearing mounting bolts	M16 class 8.8	190 N·m (140 ft·lb)
CV-3 guard fasteners	M8 class 8.8	25 N·m (18 ft·lb)
CNC chuck jaw bolts	M12 class 12.9	110 N·m (81 ft·lb)
CNC turret tool holder clamps	M10 class 12.9	65 N·m (48 ft·lb)
RW-510 torch neck	Hand tight plus 1/4 turn	-
AC-1 motor mount bolts	M16 class 8.8	190 N·m (140 ft·lb)

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12. Lubricants and Common Spare Parts

Stock code	Description	Used on
LUB-G2	NLGI 2 lithium-complex EP grease	CV-3 bearings, general
LUB-G220	ISO VG 220 synthetic gear oil	CV-3 gearbox
LUB-H46	ISO VG 46 anti-wear hydraulic oil	HP-400
LUB-W68	ISO VG 68 way oil	CNC lathes, HP-400 gibs
LUB-C500	CoolCut 500 semi-synthetic coolant	CNC lathes
LUB-RS8	Synthetic rotary screw compressor fluid	AC-1
FE-400-10	HP-400 return filter element, 10 micron	HP-400
FE-AC1-A / FE-AC1-O	AC-1 intake filter / oil filter	AC-1
FE-PW1	Parts washer solvent filter	PW-1
CT-045	MIG contact tip 0.045 in	RW-510
BRG-UCF212	Flanged pillow-block bearing, 60 mm bore	CV-3

Minimum stock: 2 CV-3 bearings, 2 HP-400 filter elements, 1 AC-1 filter kit and 50 contact tips. Reorder from the storeroom through the CMMS when stock reaches the minimum.

13. Work Orders and Records

- All PM and corrective work is recorded on a work order in the CMMS. PM work orders are generated automatically from Section 3.
- Close work orders within 24 hours of completing the job, recording readings (temperatures, pressures, oil levels), parts used and time spent.
- Equipment returned to service after guard removal or LOTO requires the supervisor's sign-off on the work order (EHS-SOP-001 Section 6).
- A PM overdue by more than 10% of its interval (for example, 25 hours for a 250-hour task) is escalated to the Maintenance Manager.

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